

Tutorial: Conjugate Heat Transfer

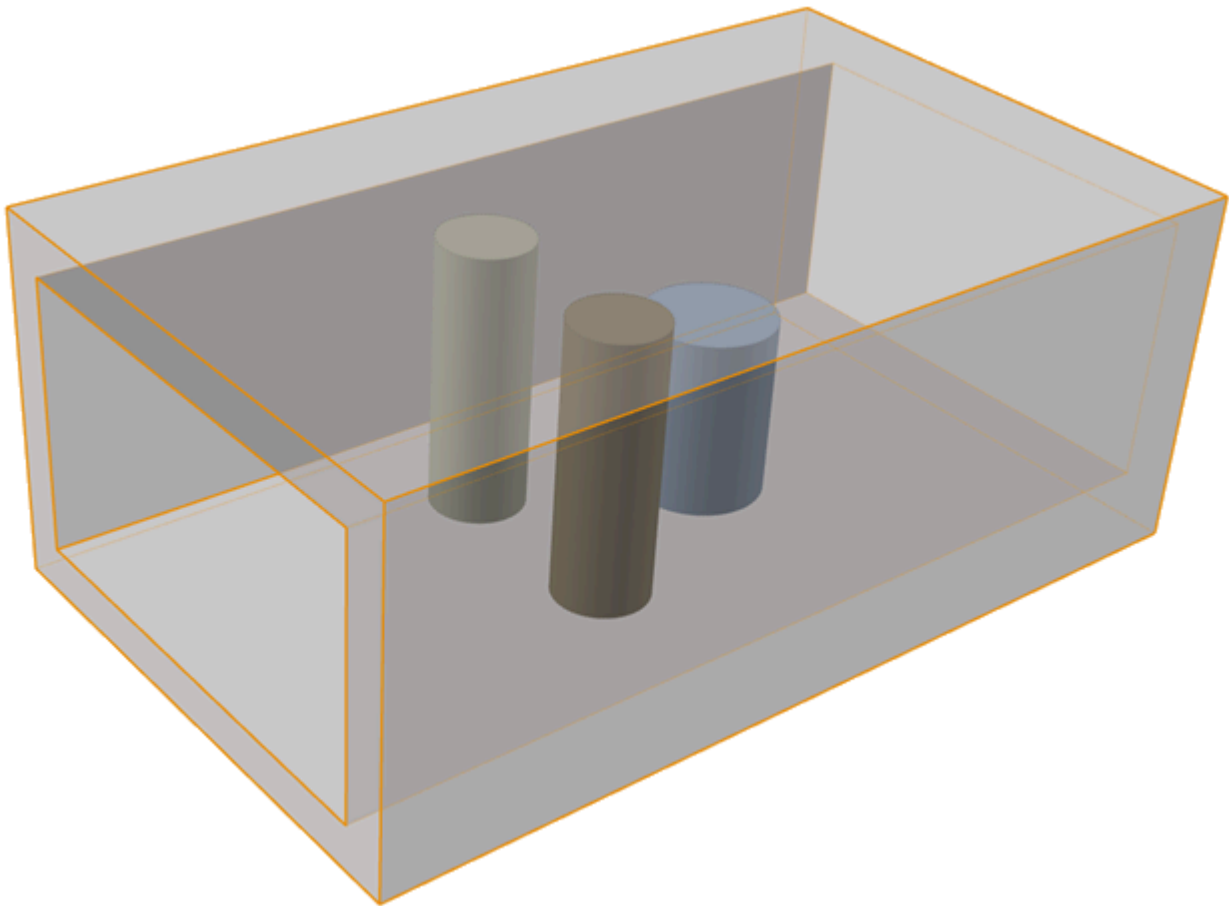
Tutorial Level: Intermediate

Learn how to run a conjugate heat transfer analysis with a simple model.

Step 1. Import Geometry

First, download and unzip this file: [CHT Tutorial.zip](#).

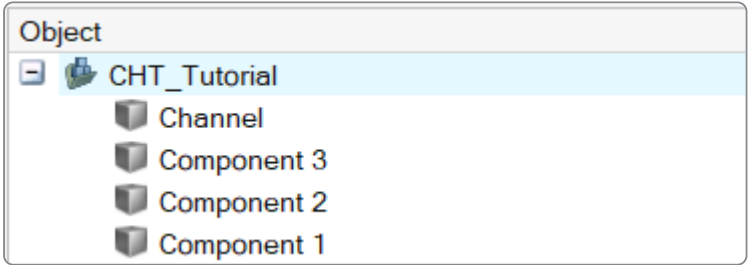
- 1. On the **File** menu, select **Import**.
- 2. Browse to the **CHT_Tutorial.x_b** file.
The model is loaded.



Note:
The model is comprised of the following parts:

- Channel
- Component 1
- Component 2
- Component 3

You can view these parts in the Object Browser:



Step 2. Designate the Bounding Solid Domain

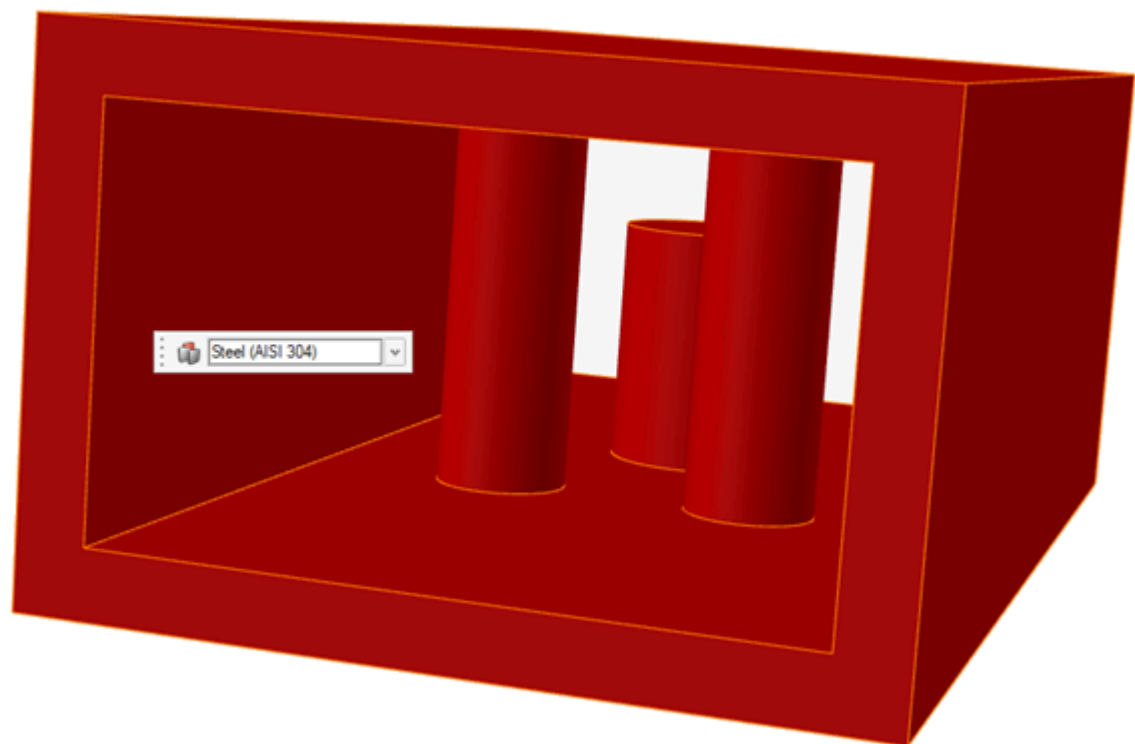
1. Click the **Bounding Solid** icon.



Tip:
To find and open a tool, press **Ctrl+F**. For more information, see [Find and Search for Tools](#).

2. Box select all parts in the model to designate the **Bounding Solid** domain.

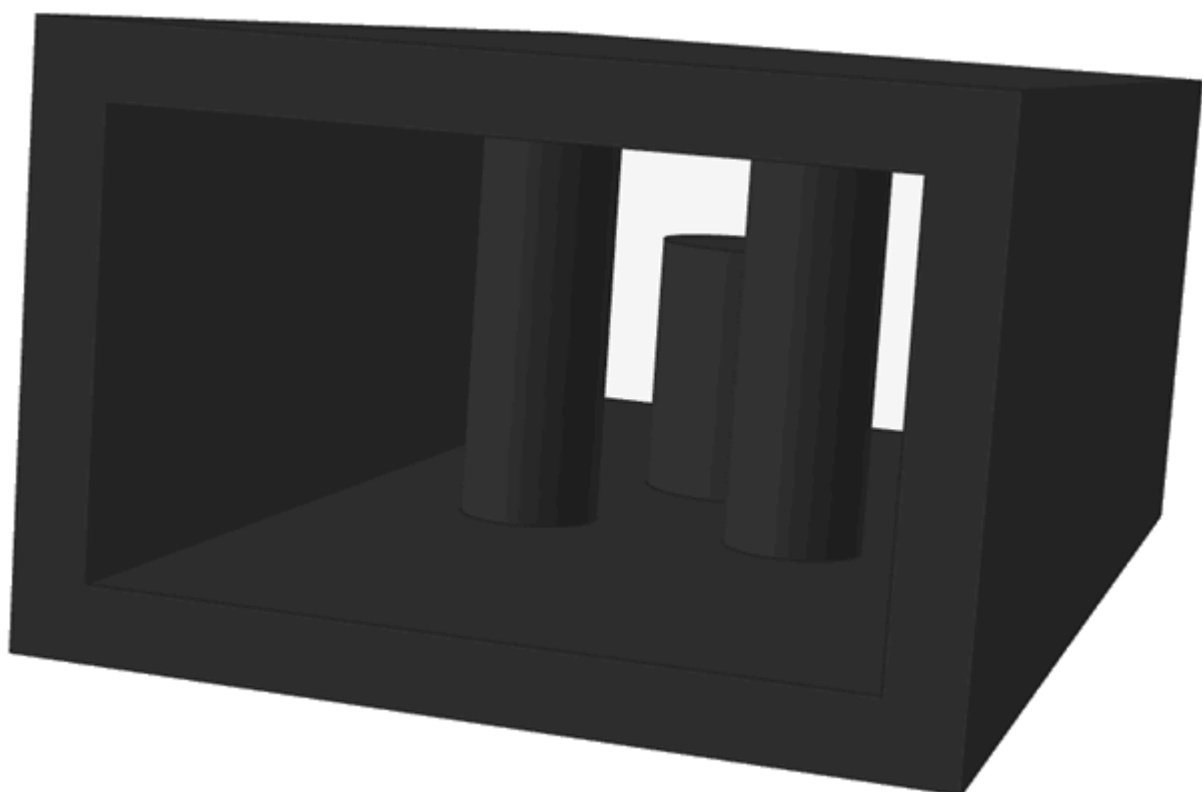
Selected parts will turn red.



3. In the microdialog, choose **Steel (AISI 304)** for the bounding solid domain.

4. Right-click and mouse through the check mark to exit, or double-right-click.

The parts designated as **Bounding Solid** domains will turn black.

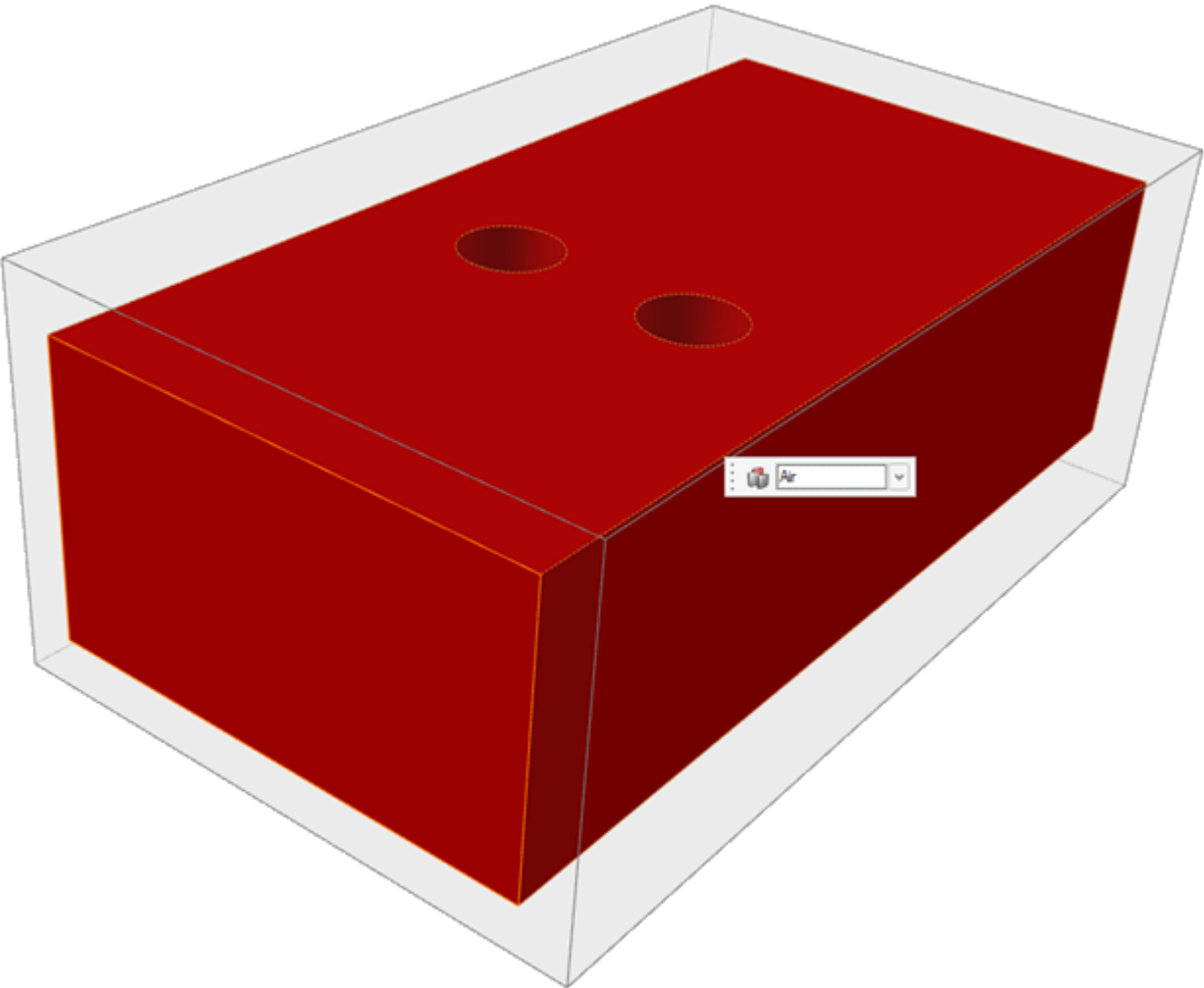


Step 3. Create the Fluid Domain

- 1. Click the **Create Fluid Domain** icon.

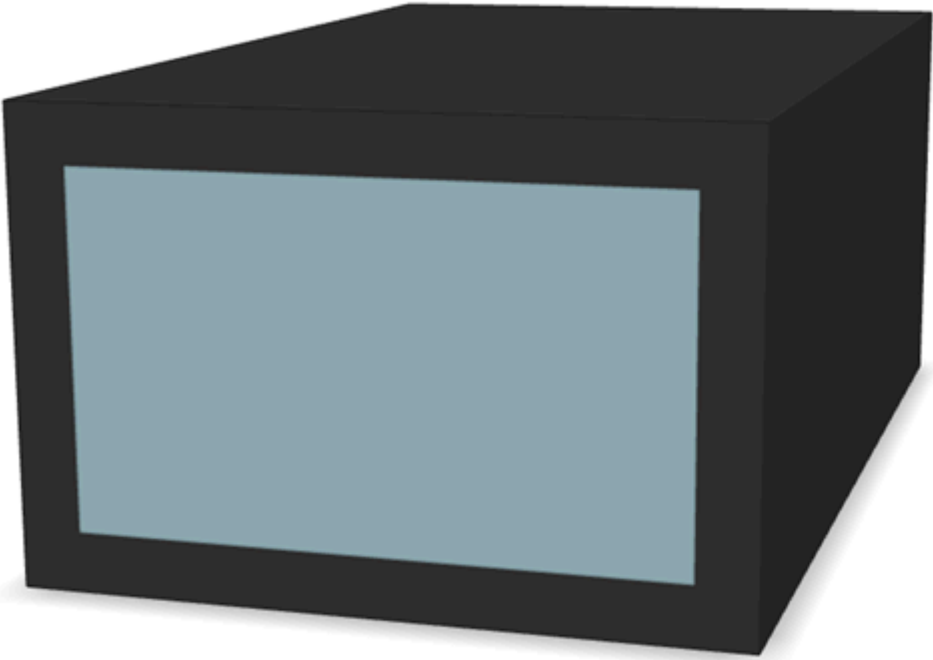


A new fluid domain is detected and created.



- 2. In the microdialog, choose a fluid type.
- 3. Right-click and mouse through the check mark to exit, or double-right-click.

The new fluid domain turns blue.

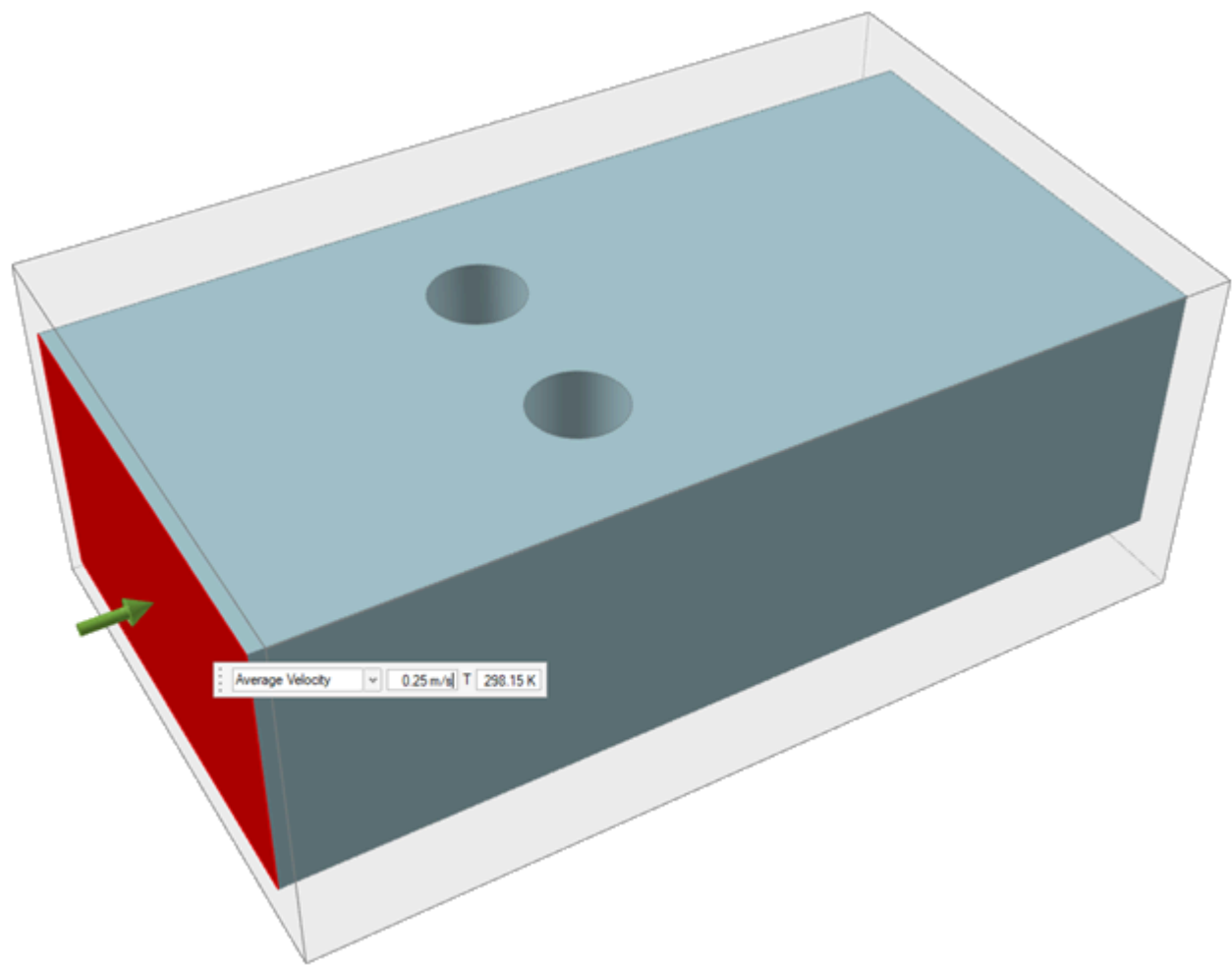


Step 4. Designate Inlets

- 1. Select the **Inlet** icon.



- 2. Select a face in the fluid domain where you would like to define an inlet. An arrow appears at each defined inlet.



- 3. In the microdialog, enter $.25 \text{ m/s}$ for the **Average Velocity**. Use the default value of 298.15 K for the temperature.
- 4. Right-click and mouse through the check mark to exit, or double-right-click.

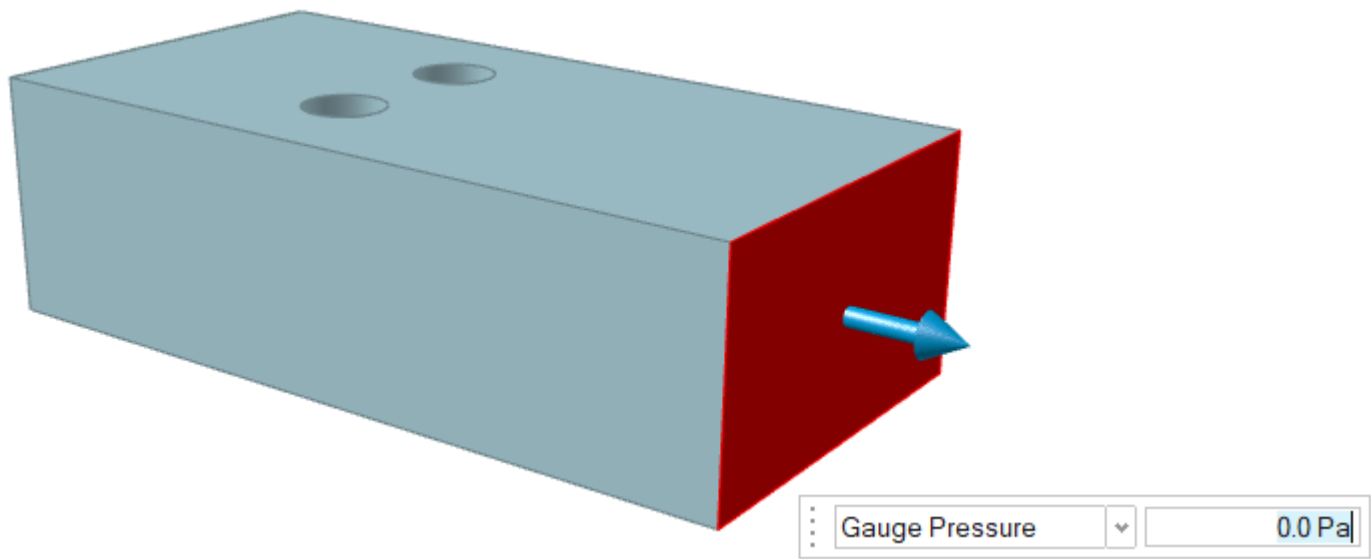
Step 5. Designate Outlets

- 1. Click the **Outlet** icon.



2. Select a face in the fluid domain where you would like to define an outlet.

An arrow appears at each defined outlet.



3. Enter 0 Pa for guage pressure for the outlet face.

4. Right-click and mouse through the check mark to exit, or double-right-click.

Step 6. Define Thermal Conditions

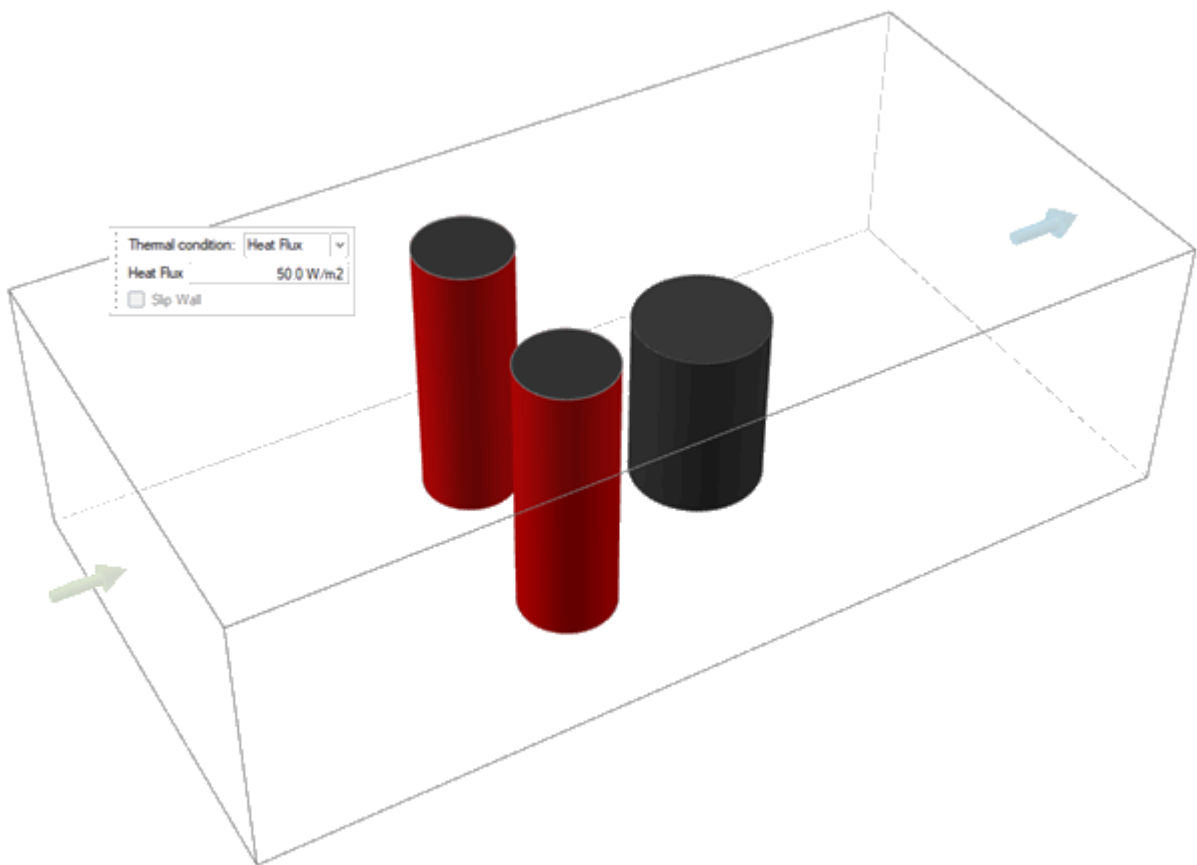
1. Click the Face icon.



2. Select Solids.

Note:
Ambient Temperature and Ambient HTC values can be changed as necessary. The default values are 293.15 K and 0 W/(m2*K) . This option does not require any selection of parts or faces.

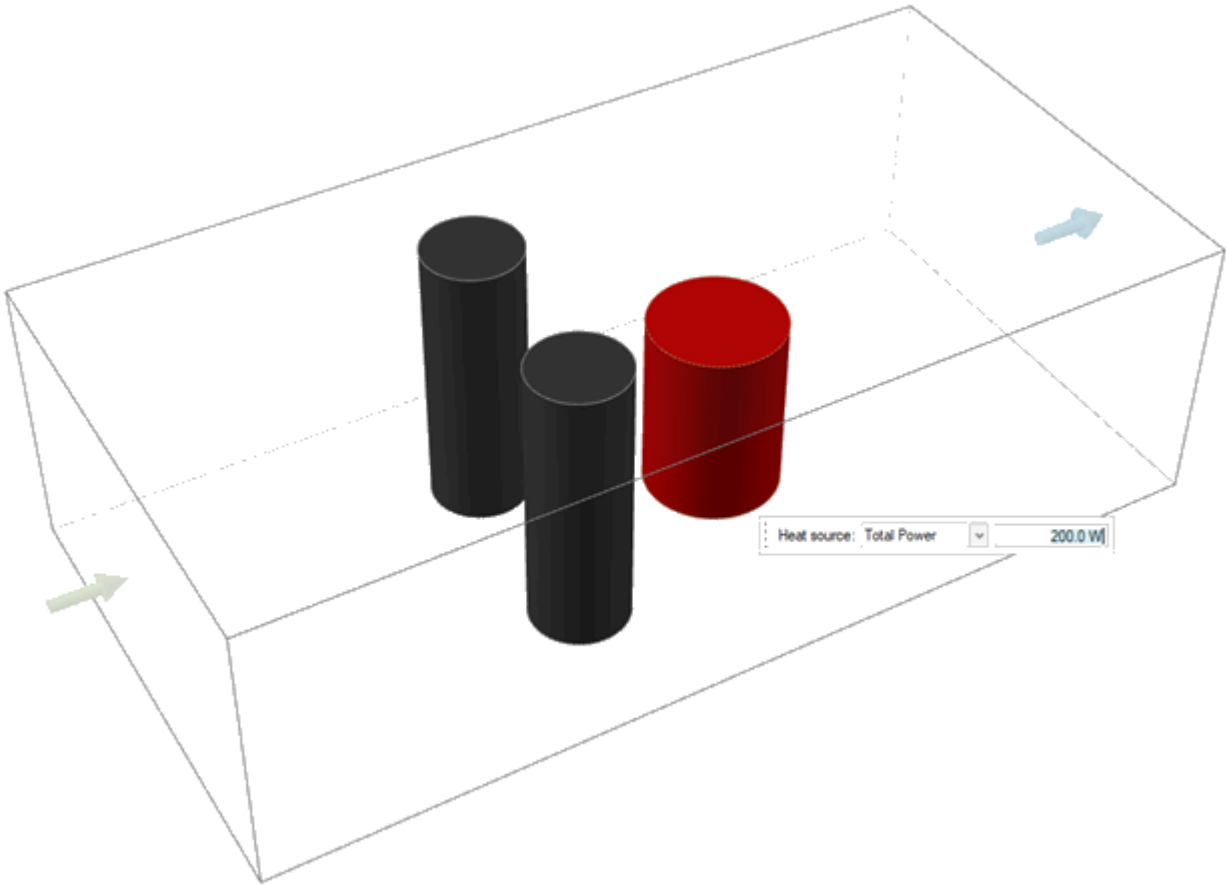
3. Select Component 1 and Component 2 in the model.



The microdialog appears with the following options:

- Temperature
- Convection
- Heat Flux

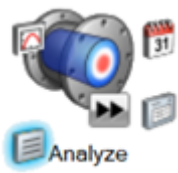
- 4. Select the dropdown menu in the microdialog, then choose **Heat Flux**.
 - 5. Enter 50 W/m^2 for the **Heat Flux**.
 - 6. On the **Fluids** ribbon, select the **Part** tool.
- 
- 7. Select **Component 3**.
 - 8. In the microdialog, select **Total Power** from the dropdown, and then enter 200.0 W .
 - 9. In the guidebar, enter 298.15 K for **Ambient Temperature**, and $25 \text{ W/(m}^2\text{K)}$ for **Ambient HTC**.



- 10. Right-click and mouse through the check mark to exit, or double-right click.

Step 7. Configure Solver Settings

- 1. Click the Custom Fluids Run icon.



- 2. In the **Simulation** tab, set the **Number of voxels** to $100,000$ and the **Voxel size** to $.0023 \text{ m}$

3. Enable **Compute thermal problem** and **Include solids**.

Run Fluids Analysis

Simulation

Advanced

Name:

CHT_TestModel

Resolution

Low

High

Voxel size:

0.0023 m

Number of voxels:

100000

Zone Refinement:

☐ Show grid

☒ Compute thermal problem

☒ Include solids

Reset

Run

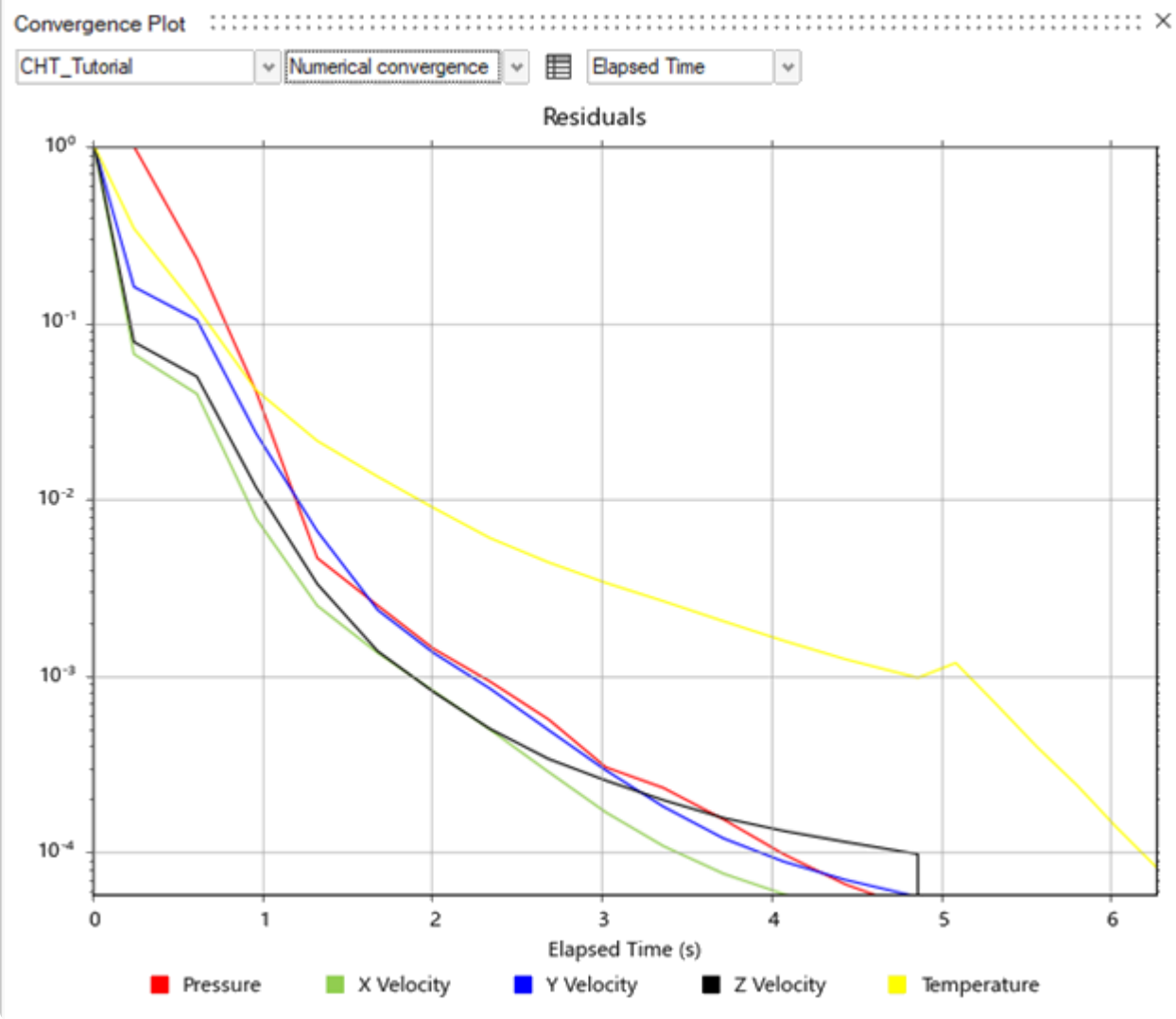
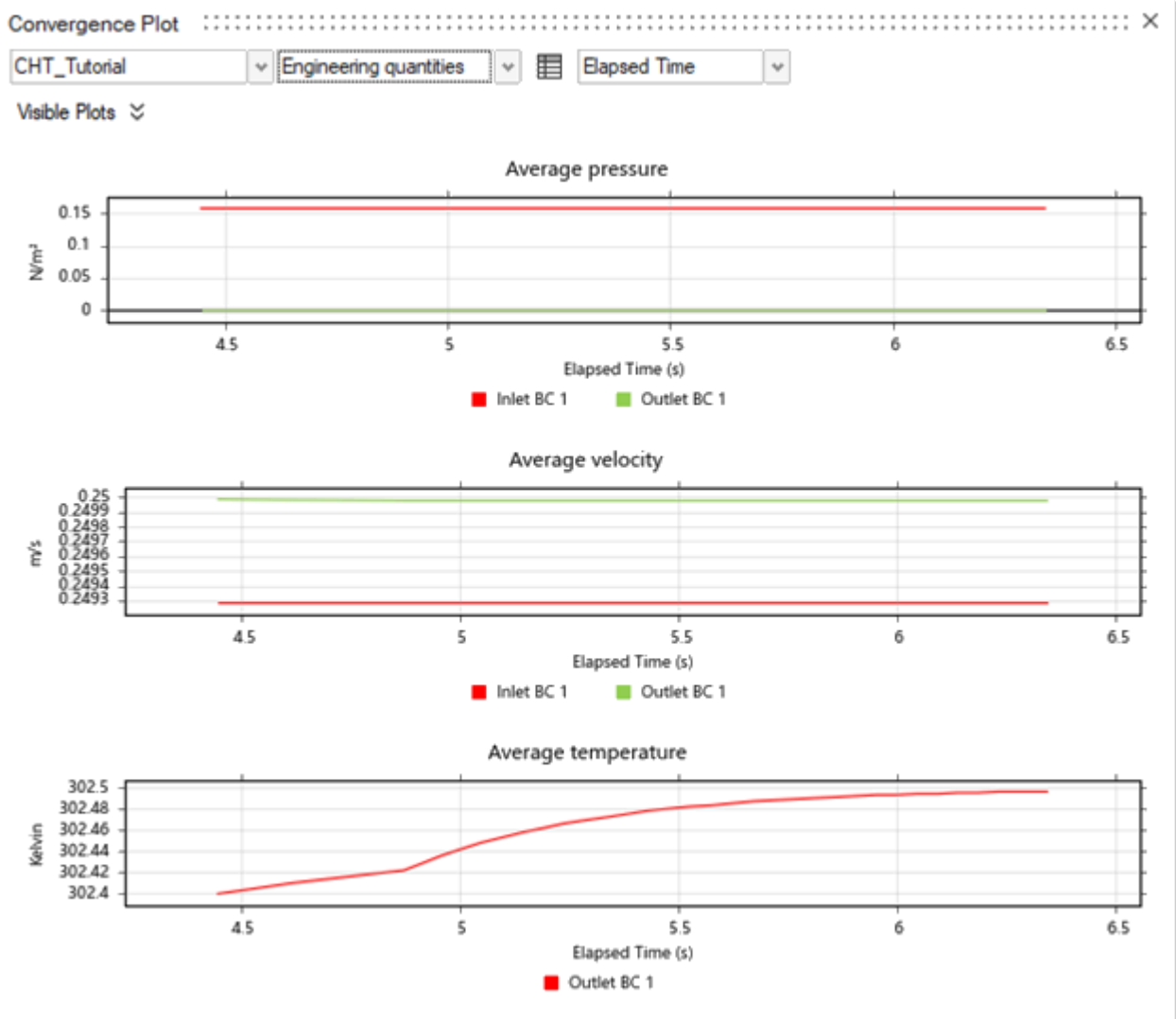
Close

4. Click **Run**.

Step 8. Monitor Convergence Plots

While the analysis is running, monitor the convergence plots. The convergence plots automatically appear after you click **Run**.

a. At the top of the **Convergence Plots** window, use the dropdown menu to visualize the simulation data with **Engineering quantities** or **Numerical convergence**.



Additionally, you can select **Show Convergence Table** to visualize the result quantities in a table.

Name	Average Pressure (N/m²)	Average Velocity (m/s)	Volumetric Flow Rate (m³/s)	Flow Fraction	Average Temperature (Kelvin)	Uniformity Velocity	Uniformity Temperature
Inlet_BC_88	0.15703	0.249284	-	-	-	-	-
Outlet_BC_90	0.0	0.250021	0.00147515	1.0	302.494	0.695389	0.422705

Step 9. Review Simulation Results

1. Once the simulation is completed, a green flag appears in the **Analyze** tool group.

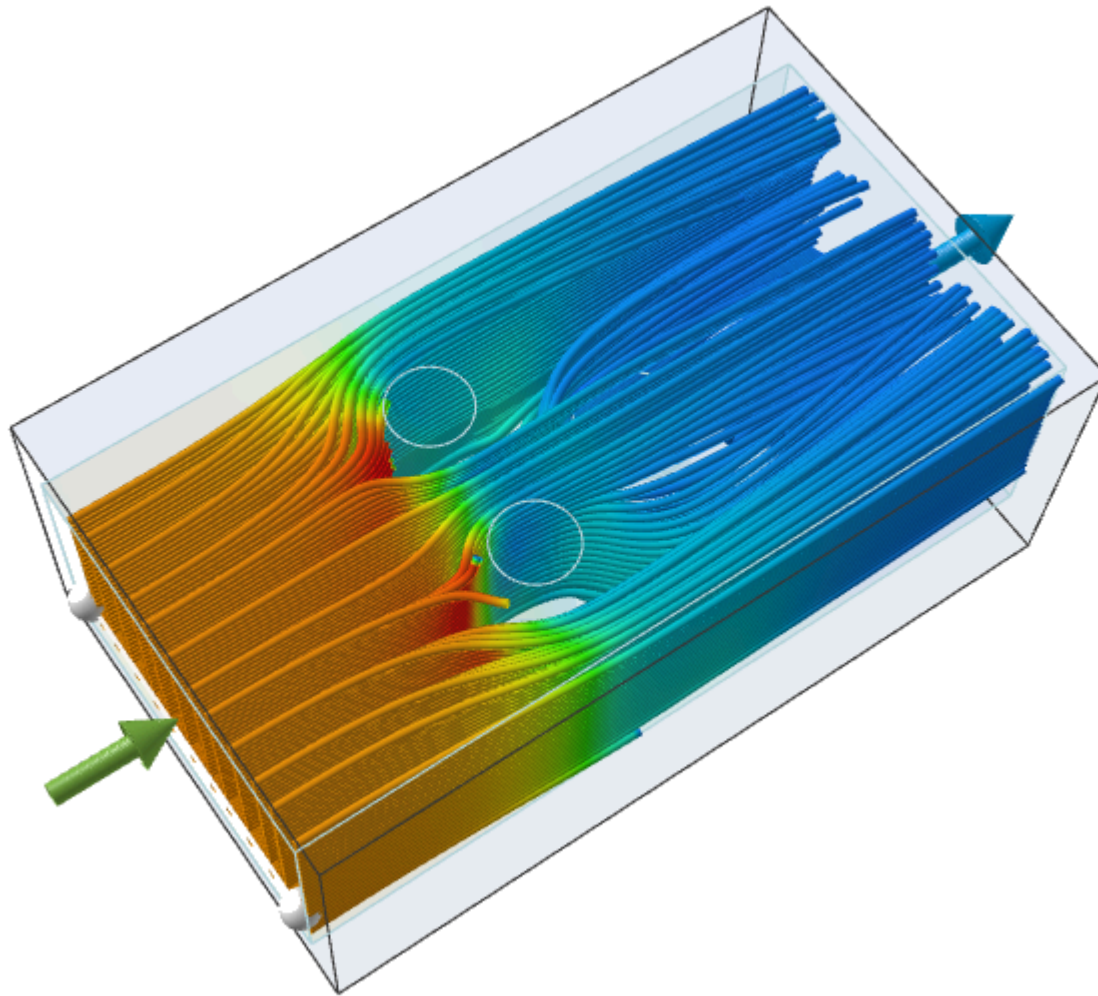


2. Click the green flag to **Import Completed Runs**.

The results from the simulation will load in the modeling window.

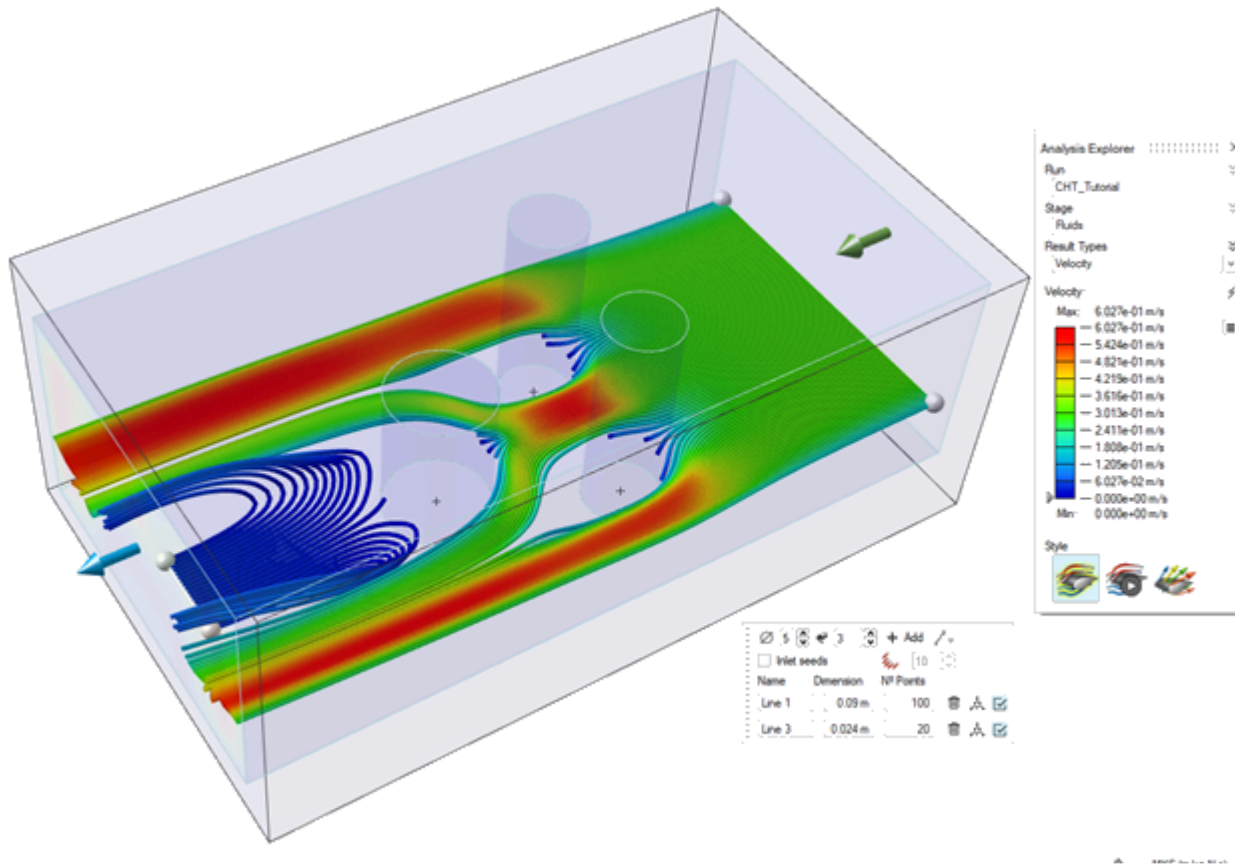
3. Generate different result types using the **Analysis Explorer**.

- a. Click  to visualize your results with streamlines.

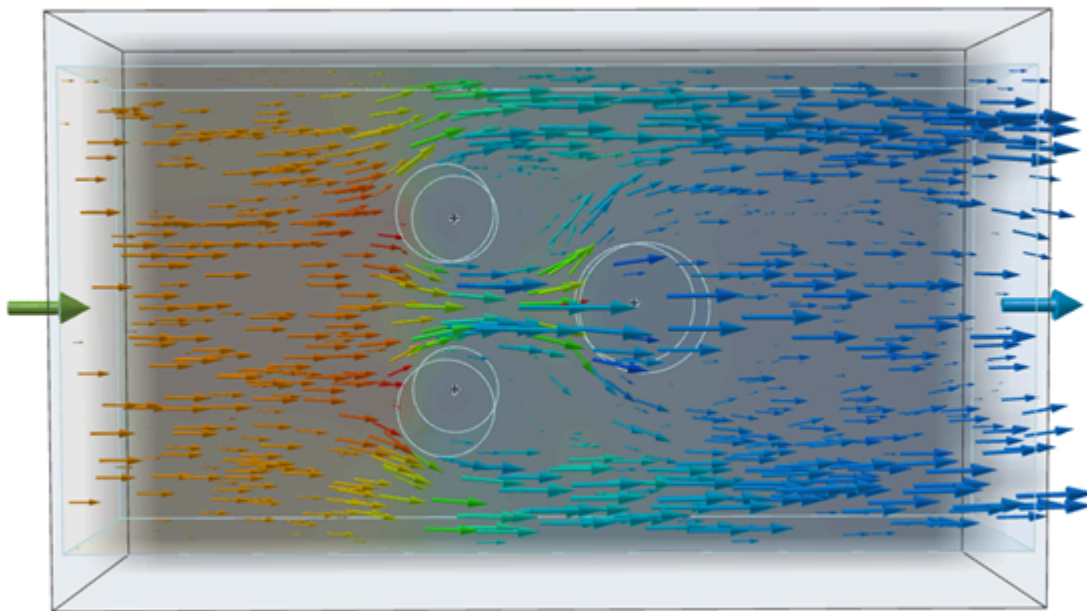


4. Optionally, you can create local streamlines in the model:

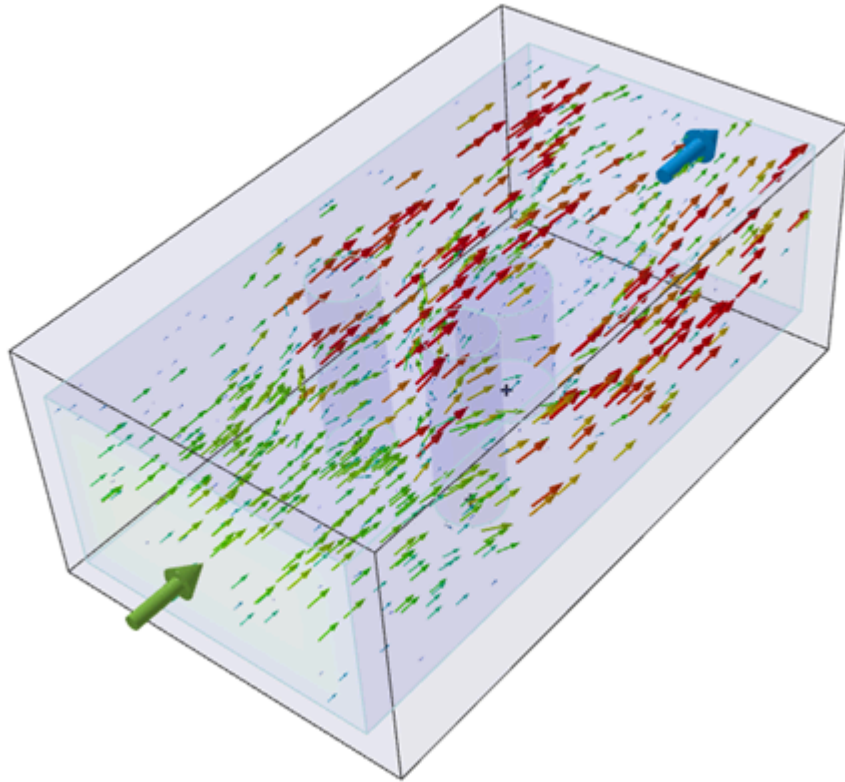
- In the streamlines microdialog, disable **Inlet seeds** to turn off inlet streamlines.
 - Click **+ Add** to create the first linear streamline region.
 - Enter a value of 0.09 under **Dimension** and 100 under **No. of Points**. Using the move arrows, reposition this linear streamline region closer to the inlet as shown in the image
 - Click **+ Add** to create the second linear streamline region.
 - Enter a value of 0.024 under **Dimension** and 20 under **No. of Points**. Using the move arrows, reposition this linear streamline region closer to the outlet as shown in the image
 - In the Analysis Explorer, change the Result Type to Velocity.
- The model should look similar to the image below:



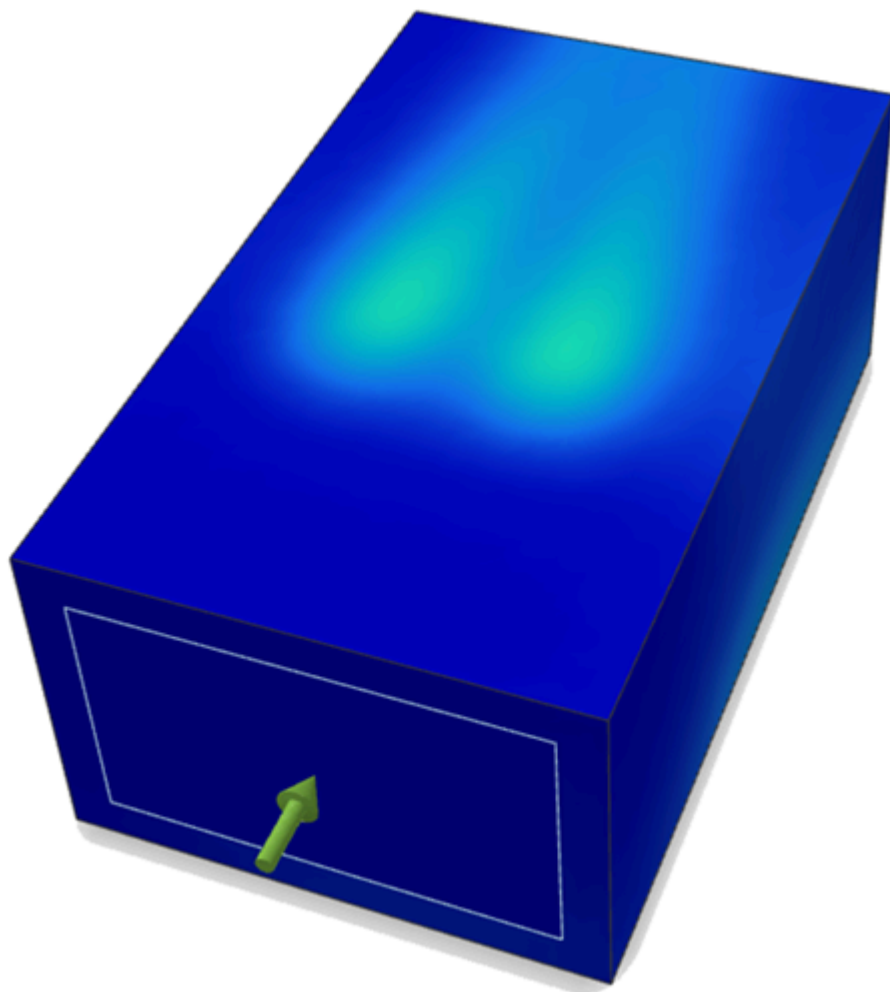
5. Click  to visualize your results with vectors.



6. Under **Result Types**, use the dropdown menu to select **Velocity**.
Results visualized as **Velocity** are displayed.

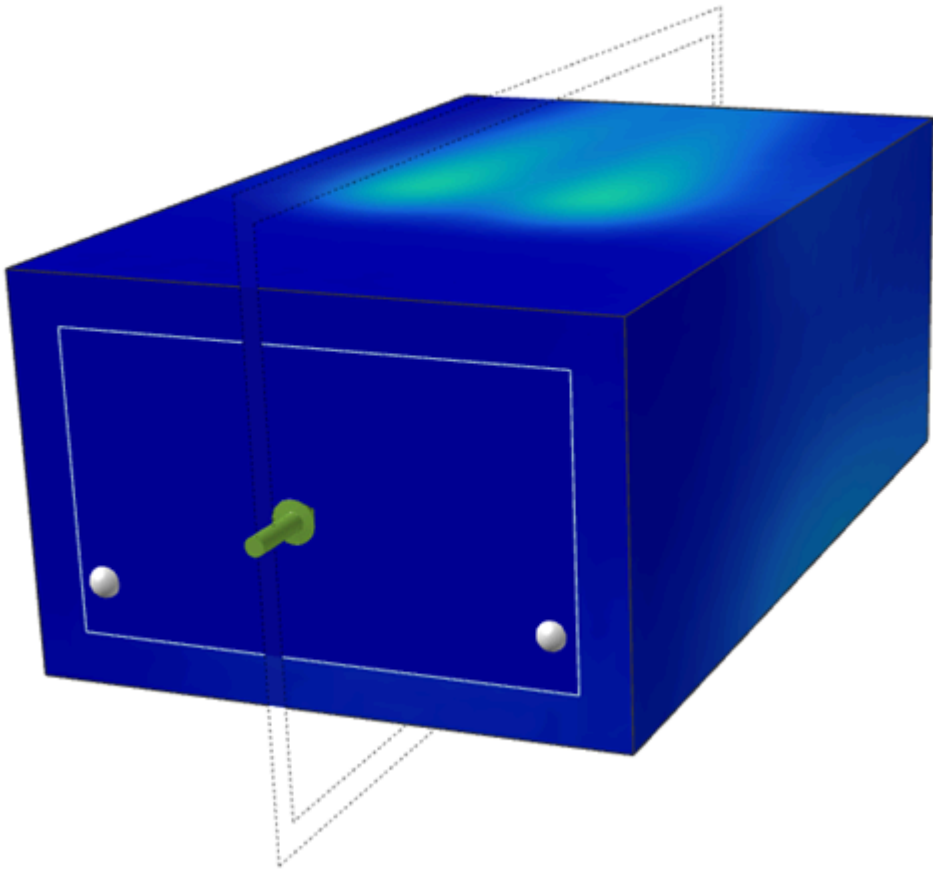


7. Deselect **Vectors** as the result Style Option.
8. Under **Result Types**, use the dropdown menu to select **Temperature**.
Results visualized as **Temperature** are displayed.

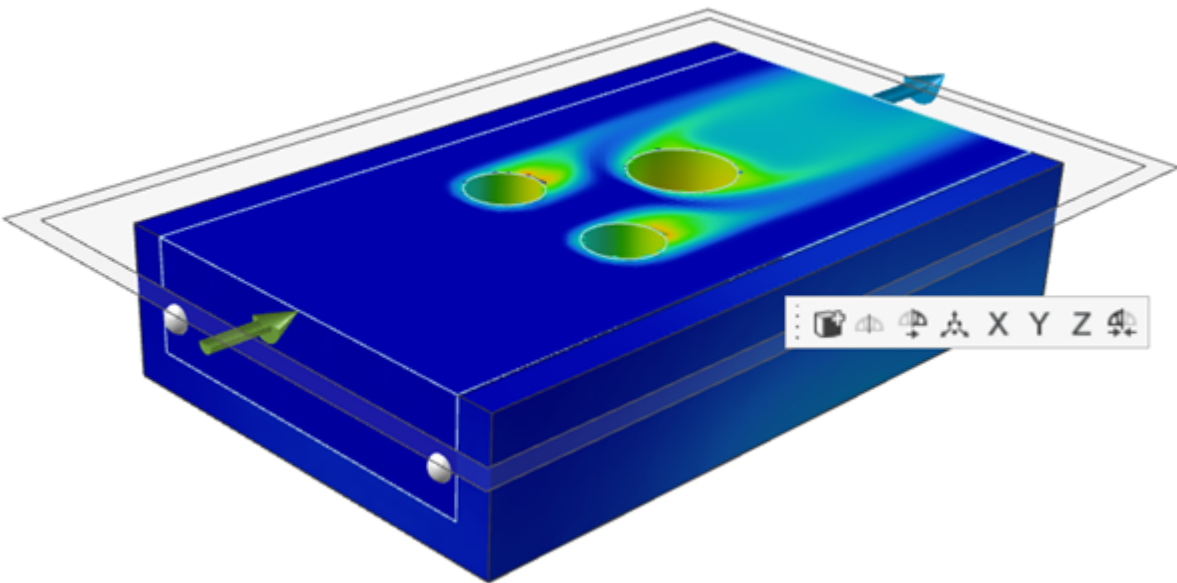


9. You can make a section cuts of the temperature results.

- a. In the Object Browser, click  to next to the component names to hide **Component 1**, **Component 2**, and **Component 3**.
- b. In the bottom-left of the modeling window, select **Section Cuts** .
- c. In the microdialog, click .
- d. Click the grey section cut pane.



- e. In the microdialog, select **Z**.
The section cut is applied.



- a. In the Object Browser, click  to next to the component names to unhide **Component 1**, **Component 2**, and **Component 3**.
The components appear in the model.

b. Right-click and mouse through the check mark to exit, or double-right click.

